LINEAR ALGEBRA · LESSON 5

Transposes, Permutations, Vector Spaces

Flip matrices, reorder rows, then step up from single vectors to whole **spaces** of vectors.

With portfolios: covariance, long-only funds and zero-cost trades.

R^TR is symmetric | Subspaces | MIT 18.06, Lecture 5

PERMUTATIONS

Same rules, different order

A permutation only **re-lists the equations.** Same facts, same answer.

$$\begin{aligned} y &= 3 \\ x + y &= 8 \end{aligned} \quad \xrightarrow{\text{swap}} \quad \begin{aligned} x + y &= 8 \\ y &= 3 \end{aligned}$$

Left: the first rule has no x, a 0 where the pivot should be, so elimination is stuck. Right: already triangular: $y = 3$, $x = 5$.

RULES

$PA = LU$: reorder first, then eliminate. Undo a reorder with P^T: $P^T P = I$. 3 rules $\rightarrow$ 6 orders (3!).

Rows become columns

$(A^T)_{ij} = A_{ji}$. In A each column is a fund; in A^T each row is a fund's profile:

$$\begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 12 & 1 \\ 1 & 7 & 0 \end{bmatrix}$$

Portfolio return as a row times a column:

$$r^T x = \begin{bmatrix} 12 & 7 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = 60 + 21 + 30 = 111$$

A RULE

$(AB)^T = B^T A^T$

Transpose each, reverse the order. For two clients:

$$(AX)^T = \begin{bmatrix} 10 & 105 \\ 10 & 90 \end{bmatrix} = X^T A^T$$

AX had one column per client; its transpose has one **row** per client. Same report, sorted the other way.

R^TR is always symmetric

$(R^TR)^T = R^T(R^T)^T = R^TR$. Scenario returns of Nifty and bonds:

$$R = \begin{bmatrix} -10 & 4 \\ 10 & 7 \\ 30 & 10 \end{bmatrix} \Rightarrow R^TR = \begin{bmatrix} 1100 & 330 \\ 330 & 165 \end{bmatrix}$$

Entry $(1,2)$ and $(2,1)$ are the same dot product: $\text{Nifty} \cdot \text{bonds} = -40 + 70 + 300$.

Why covariance is symmetric

Subtract the average returns (10% and 7%), then $\frac{1}{3}C^TC$:

$$\text{Cov} = \begin{bmatrix} 266.7 & 40 \\ 40 & 6 \end{bmatrix}$$

A covariance matrix is C^TC in disguise, so it is **always symmetric**: cov(Nifty, bonds) = cov(bonds, Nifty).

Add and scale without leaving

A vector space is closed under **adding** and **multiplying by any number**, and contains 0.

- $(7, 3) + (4, 6) = (11, 9)$: merge two books
- $-1 \times (7, 3) = (-7, -3)$: short the lot
- $0 \times (7, 3) = (0, 0)$: the empty portfolio

All portfolios (long or short) form the space $\mathbb{R}^2$.

NOT A SPACE

Long-only portfolios fail

$x \geq 0, y \geq 0$: adding is fine, but $-1 \times (7, 3) = (-7, -3)$ escapes.

Fixed budget $x + y = 10$ fails too: $(7, 3) + (6, 4)$ has budget 20, and 0 isn't in it.

WHY IT MATTERS

A long-only constraint breaks linearity: portfolio construction needs inequality-constrained optimisation.

SUBSPACES

Zero-cost trades are a subspace

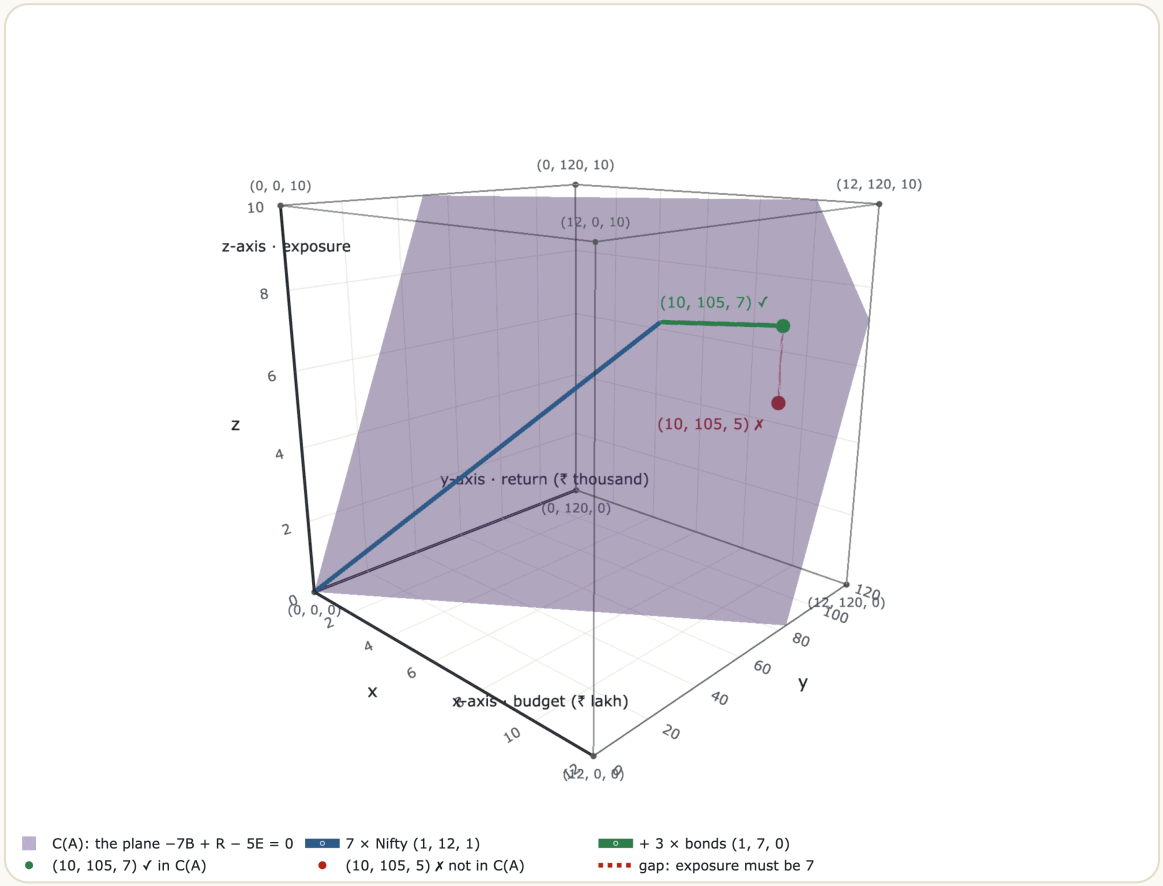
Trades whose changes add to zero: $x + y = 0$. Sums and multiples stay zero-cost, and 0 is in it: **a line through the origin.**

Subspaces of $\mathbb{R}^3$: $\{0\}$, lines through 0, planes through 0, all of $\mathbb{R}^3$.

Zero-cost **and** zero-exposure: the line through $(-6, 1, 5)$, a trade that changes only the return (+10).

C(A): every target you can reach

All combinations of the columns. Nifty + bonds with three rules gives a **plane** in $\mathbb{R}^3$:



$(10, 105, 7)$ ✓ reachable; $(10, 105, 5)$ ✗: exposure always equals the Nifty amount.

RECAP

Key takeaways

1. $PA = LU$; $n!$ permutations; $P^{-1} = P^T$.
2. $(A^T)_{ij} = A_{ji}$; $(AB)^T = B^T A^T$.
3. $R^T R$ is symmetric: so is covariance.
4. Spaces are closed under $+$ and scaling.
5. Long-only isn't a space; zero-cost trades are.
6. $C(A)$ = targets the columns can reach.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A column-space explorer for your own instruments and targets, 3D plots and worked hedging and options examples:

learn.krisarj.com

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